



## ARI-NextGen Student Fellowship Proposal

- What is the title of your project?

### Blockchain Enabled Agricultural Data System with AI for Swine and Soybean Production

- Identify the Research Problem (2500 character limit)

Smallholder swine farmers and soybean growers face systemic challenges in recording, verifying, and using their production data. For swine producers, critical records such as breeding histories, feed intake, growth rates, veterinary treatments, and slaughter weights are often paper based or stored in databases controlled by large integrators, which limits transparency, reduces bargaining power in auction markets, and complicates access to insurance or credit. Similarly, soybean producers may control essential crop records, such as harvest date, yield per acre, moisture content, and machine logs, but these records are typically not traceable or verifiable by multiple stakeholders. The lack of verifiability and shared transparency affects both swine and soybean producers, making it difficult for farmers to contest pricing disputes or claim insurance payouts.

From a regulatory perspective, the absence of verifiable, on-chain records complicates disaster relief and emergency insurance distribution. After floods, droughts, or disease outbreaks, agencies struggle to confirm production losses when farm-level data is fragmented or unverifiable, delaying financial support to farmers.

Blockchain technology provides an **immutable and traceable ledger**, ensuring that once production records are entered, they cannot be altered or deleted, and every transaction can be

independently verified by stakeholders. Recently, blockchain technology has seen widespread adoption across various domains. For instance, the U.S. Department of Commerce recorded the 2025 second-quarter Gross Domestic Product (GDP) on the blockchain [1]. **This research proposes an open-source blockchain system that allows farmers to VALIDATELY record and share swine and soybean production data.** By placing these records on-chain, the project will enable fairer pricing, timely access to insurance and relief assistance, improved traceability, and stronger participation of smallholder farmers in cooperative, national, or international markets.

- What is the significance of this problem? (2500 character limit)

Livestock and crop production are the foundation of food security and rural economies in the United States. Pigs and soybeans are cornerstones of both domestic consumption and international trade. However, the inability of smallholder farmers and cooperatives to maintain transparent, verifiable records puts them at a structural disadvantage.

For pig producers, incomplete or inaccessible data on breeding, feed, health interventions, and slaughter weights reduces their credibility in the market and limits their ability to negotiate fair prices. This opacity also undermines farmers' access to credit and insurance, which increasingly require accurate performance and health data to assess risk.

Soybean growers face a parallel challenge. Disputes over yield per acre, harvest dates, or moisture content often result in contested payments at grain elevators. In the absence of traceable data, producers may be forced to accept unfavorable terms. Additionally, during extreme events such as droughts, floods, or pest outbreaks, the lack of trustworthy crop records delays the verification process for insurance and disaster relief programs, leaving farmers financially exposed at precisely the moment when assistance is most critical.

From a broader economic perspective, the lack of trustworthy agricultural data creates inefficiencies throughout the supply chain. Buyers face uncertainty in product quality and volume, insurers and regulators spend resources on verification and fraud prevention, and farmers absorb unnecessary financial risk. These inefficiencies not only erode trust but also suppress the competitiveness of smallholder farmers in markets dominated by large-scale agribusinesses.

An open-source blockchain system addresses these barriers by giving farmers direct ownership of their production records and creating a tamper-proof ledger accessible to cooperatives, insurers, buyers, and regulators. By ensuring that pig and soybean data are transparent, auditable, and permanent, the system reduces disputes, speeds up insurance claims and disaster assistance, and creates a more balanced and resilient agricultural economy.

- What is known about this problem? Use primary literature sources to support your claims (add references to the REFERENCES section, last box at the end of the application). What new information will your research add? (3000 character limit)

Accurate recordkeeping is essential in agriculture to ensure fair access to benefits, insurance, and disaster relief. In swine production, detailed records of breeding cycles, disease events, feed conversion ratios, and sale weights are critical for herd management, traceability, and profitability [2]. Similarly, in soybean farming, accurate documentation of harvest date, yield per acre, moisture content, and machinery use is central to determining farmer income, crop insurance payouts, and eligibility for relief programs [3]. However, these data are often fragmented across multiple systems or maintained in paper records, making them vulnerable to loss, manipulation, or misreporting. **This creates challenges for market transparency, slows verification of insurance claims, and limits regulators' ability to quickly assess disaster-related losses.**

Fraud cases highlight the real-world consequences of unverifiable data. For example, claimants submitted false applications for financial relief under USDA benefit programs, and investigations revealed that many claimants had not suffered discrimination and, in most cases, had not attempted to farm. **The scheme involved 192 claims, resulting in over \$11.5 million in fraudulent payments, demonstrating how lack of verifiable farm level data can be exploited [5].**

Existing blockchain initiatives in agriculture are predominantly pilot projects developed by private enterprises. For example, IBM has implemented blockchain applications in food supply chains to enhance traceability and settlement processes. **However, these platforms are not fully open-source or easily deployable for widespread use [4].**

Research prototypes integrating blockchain with agriculture, including livestock traceability and farm data management, often **lack full deployment pipelines, scalability, and user-friendly interfaces**, limiting real world adoption. For example, one study demonstrated a distributed ledger and smart contracts to improve traceability and involve small farmers, but noted difficulty in framework accessibility and deployability [4].

This situation highlights the need for open-source, user-friendly blockchain systems that are easily deployable in agricultural settings. The proposed research will leverage large language models (LLMs) and AI agents to allow voice-based data entry and checking, enabling farmers who may have limited technical skills to interact with a complete blockchain database system. The AI agent will review inputs for significant errors before logging them on-chain, ensuring accuracy, reliability.

**This research will contribute by developing an open-source, easily deployable blockchain system for pigs and soybeans, integrating AI-assisted voice-based input and AI-assisted data validation.** By ensuring data integrity and on-chain traceability, this system aims to improve verification of farm data, reduce disputes, and support equitable access to agricultural benefits and relief programs.





- What is your experimental approach? Describe the dataset(s) you expect to generate, and what your analyses will seek to answer. (5000 character limit)

1. **Blockchain Layer (Solana):** The system will use the Solana blockchain as the underlying layer to record all agricultural data securely and immutably.
2. **Smart Contract Framework (Anchor):** Smart contracts will be developed using Anchor, a Rust-based framework that simplifies Solana program development, testing, and deployment.
3. **Blockchain Client SDK (@solana/web3.js):** The web or mobile application will use the @solana/web3.js JavaScript SDK to interact with Solana smart contracts and submit transactions.
4. **AI / Large Language Model Integration (GPT):** OpenAI's GPT will process and structure voice input from farmers, converting it into structured agricultural records and checking for inconsistencies or errors before storing the data on-chain.
5. **Voice-to-Text Processing (Whisper):** OpenAI's Whisper will convert farmer voice input into text that GPT can interpret and validate.
6. **Back-End / Middleware (Node.js / Express):** A Node.js server with Express will handle API calls between the front-end interface, GPT/Whisper, and the Solana blockchain.
7. **Database (PostgreSQL / SQLite, optional):** An optional off-chain database can be used for temporary storage or caching of data before it is committed to the blockchain.
8. **Front-End / Farmer Interface (React / React Native):** A web or mobile front-end built with React or React Native will provide a user-friendly interface for farmers, supporting voice input, easy navigation, and data review.
9. **Wallet Integration (Phantom Wallet SDK):** Farmers will authenticate and authorize blockchain transactions using a wallet integration such as Phantom Wallet.
10. **Deployment / Testing:** The system will be tested on the Solana Devnet, and once stable, deployed to the mainnet for real-world use.
11. **Historical Data Testing:** We will collaborate with the **Precision and Automated Agriculture Lab at the University of Missouri, led by Dr. Jian Feng Zhou**, to test the

system using historical swine and soybean data.

12. **Pilot Testing with Farmers and Stakeholders:** Smallholder farmers, insurance companies, and USDA regulators will participate in pilot testing to evaluate usability, accuracy, and compliance with regulatory standards.
13. **Survey and Analysis:** The pilot will include survey-based analyses of farmer experience, system usability, and stakeholder feedback to assess adoption potential and effectiveness.
14. **Dissemination:** Findings and demonstrations of the blockchain system will be presented at industry and academic conferences, including the **US Conference on Precision Livestock Farming** and the **US Soy Global Trade Exchange & Specialty Grains Conference**.

- REFERENCES: Please list your references here. Use a citation style consistent with a journal appropriate for your field of study (e.g., HortScience, Crop Science, Journal of Wildlife Management, Theoretical and Applied Genetics, etc.) (10,000 character limit)

1. U.S. Department of Commerce, 2025. Department of Commerce posts 2nd Quarter Gross Domestic Product to the blockchain  
<https://www.commerce.gov/news/press-releases/2025/08/department-commerce-posts-2nd-quarter-gross-domestic-product-blockchain>
2. Resti Y et al., 2024. *A review of on-farm recording tools for smallholder dairy farming*. Tropical Animal Health and Production 56: 168.
3. UNDP, 2021. *Blockchain for Agri-Food Traceability*. United Nations Development Programme.
4. Patel AS et al., 2023. *Blockchain technology in food safety and traceability concerning livestock products*. Heliyon 9(6): e16526.
5. U.S. Department of Justice, 2021. *Seven defendants sentenced to federal prison in \$11.5 million fraud case involving Black, Hispanic, and women farmers*.  
<https://www.justice.gov/usao-edar/pr/seven-defendants-sentenced-federal-prison-115-million-fraud-case>

